Summer of Code 2026

Infinite World Generator

Report

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Week 1

This week wasn't really about learning a lot of new theory. Three tutorials to go through and get comfortable with the basic tools we'd be using for the project.

The first tutorial was about generating natural-looking terrain using geometry nodes. It mainly used a color ramp with a noise texture to shape the terrain, and used slope to mask where grass texture and rock texture should show up.

The second one was a beginner geometry nodes tutorial that walked through making a fractal 2D sponge (a Menger sponge). I followed along and remade it, and then went a step further and made a 3D version of it too.

The third video was about Perlin noise. Explaining what it actually is and why it's used so widely in procedural generation.

For the implementation this week, I just followed what the first tutorial covered. Generating terrain in geometry nodes using a color ramp driven noise texture, and using slope to mask grass and rock textures.

Figure 1: 1st week mini project

Week 2

This week was about using layered noise for better terrain, and also about generating a desert biome that actually looks natural using Gabor and Voronoi textures. The main focus for the week, though, was biome generation using a height mask.

I ended up going down two different paths for this. First, I generated terrain using close to 5 layered noise textures and used a height map to blend biomes. Snow at the top, forest in the mid levels, and desert at the low, flat areas, with the blending done through color (white for snow, green for forest, orange for desert).

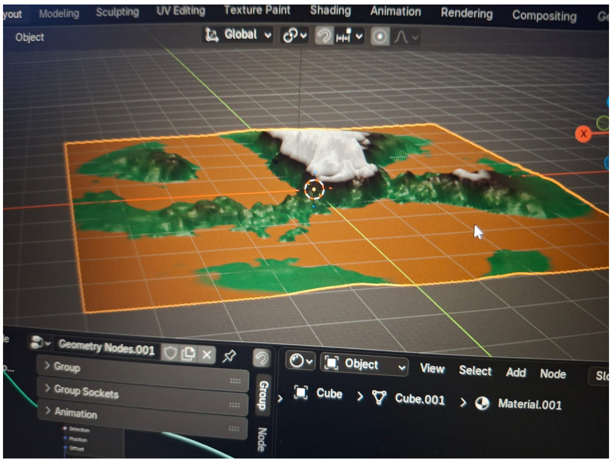

Figure 2: First version of week 2 mini project

But then I saw some discussion happening in the group about just placing three different terrains side by side and blending between them instead of relying purely on a height map. That approach seemed like it might work better, so I switched over and spent a lot of time on it, since getting the texture blending and asset placement to look natural took a lot of care.

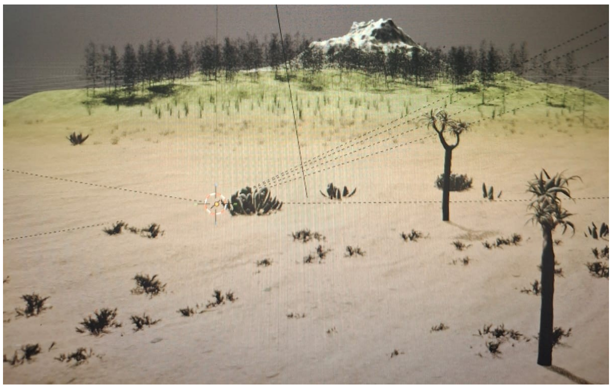

Figure 3: 2nd version

Eventually the mentor clarified that he specifically wanted us to use the height map approach. So I dropped the side-by-side idea, went back to the original height-map method, and finished the week's mini project fairly quickly after that.

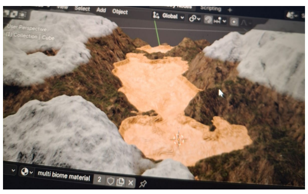

Figure 4: Final version

Week 3

This week was about learning how to scatter objects using weight paint, using nodes like Distribute Points on Faces and similar ones. There was also a video explaining Poisson disk sampling, another one on Voronoi regions, and finally an hour-long video on river generation, covering how to make a river flow and how to manually draw bezier curves to generate one.

For the mini project, I built on the terrain from before using layered noise again, but this time scattered plants, rocks, and grass using different weight paints. I also manually drew a bezier curve and placed a river along it. It didn't turn out looking the best, but the mentor mentioned that the main thing to take away from this week was really just the weight paint part, so that was fine.

Week 4

This week was about road generation and learning the A* algorithm, along with some videos on scene polishing, atmosphere and lighting, and camera animation.

For the mini project, I started with a plane and placed a big red hut near the center to act as the village center. I then scattered a few smaller red huts around it using Distribute Points on Faces with Poisson disk sampling, and labeled these as the important locations of the village. I used shortest paths to connect the village center to each of these locations, then gave the paths a curve profile to make them 3D, and added some noise to make the roads slightly curvy instead of dead straight.

The most interesting part was using these roads almost like a weight paint to scatter yellow huts, which I used as houses, more of them close to the roads, fewer further away. The result wasn't great looking, though, mainly because the huts were imported assets with a lot of vertices, so I couldn't scale them down and add more without Blender crashing. And then finally added some camera animation.

Week 5

This week was about introducing Python into Blender. I watched a few YouTube videos to get familiar with the bpy module/library.

For the implementation, I first made a terrain using geometry nodes, keeping a few parameters open as inputs, which are seed, mountain strength, terrain scale, tree density, and tree seed. Then I wrote a Python script that first creates a tree object by joining a cylinder and an icosphere, and assigns materials to them before joining. The basic idea was to use the random library to generate values within ranges I defined, to make the terrain look good, and assign those values to the input sockets. Running the script once doesn't directly give trees though. After running it, I had to go into the geometry node editor and manually select the tree object in the Object Info node for it to actually distribute.

Week 6

This week was about learning a bunch of algorithms used in infinite world generation. The first idea was understanding how keeping the same seed value gives the same terrain every time, like how noise generated using a seed value along with x and y coordinates gives the same height value at a particular (x, y) even if visited a hundred times.

I then learned about chunk-based generation, and watched videos explaining cellular automata, as well as flood fill, L-systems, Wave Function Collapse, and Binary Space Partitioning, just to understand the concepts behind each.

For the implementation, I built a Python script that generates a 9-chunk terrain that looks like a single continuous chunk, since the noise is based on x, y, and the seed value. I first wrote a function that generates a chunk given its world coordinates, and also wrote my own noise function meant to replicate how fBM noise works in geometry nodes. I then looped over all the vertices and assigned z = noise(x, y, seed).

The result initially looked like mountains and inverted mountains everywhere, so I added a fix where if height < 0, then height = height * 0.05, which smoothed things out. I also created two materials at the start, for snow and rock, and used slope to apply the textures. Finally, I stored information about each chunk and its average height into a JSON file, as well as in a Python dictionary.

Week 7

This was the last week, and it was about learning how to make our own Blender add-on, things like how the folder/zip structure needs to be set up, what each file is responsible for, and general Blender add-on conventions. I learned most of this through a mix of googling and using AI, relying on AI a lot more this week than in previous weeks to understand what goes into each file.

For the implementation, I made a zip file containing 5 Python files and one assets folder.

The 5 files were `__init__`, `generator`, `panel`, `properties`, and `operators`, and I went through understanding what each of these is meant to do.

The add-on itself adds a WorldGen tab in the sidebar, with four parameters: Seed, Terrain Size, Mountain Strength, and Tree Density, self-explanatory stuff. Below that are two buttons, Generate World and Randomize Seed, which do what their names suggest. One thing to note is that Randomize Seed doesn't automatically regenerate the terrain. Generate World has to be clicked again after randomizing.

This version isn't chunk-based, it just generates a single chunk. Making it generate 9 chunks would have been easy enough, but that wasn't part of what was asked for in this week's mini project description, which is also the final project. So overall, this week was mainly about learning add-on structure and conventions and applying that to package everything into a working add-on.